

positioning, speed regulation, and motor behaviour, making it a practical reference for developing robotic arms, actuators, and mobile robots.

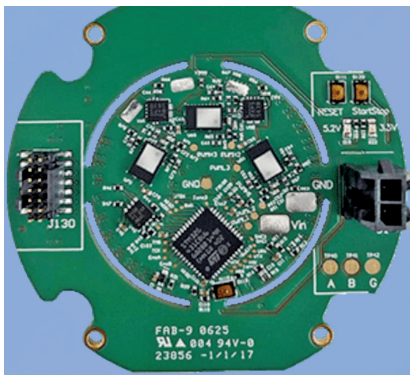
Applications. This design is suitable for applications requiring high-precision position and speed control.

Link. <https://www.electronicsforu.com/electronics-projects/compact-robotics-and-automation-reference-design-kit>

OEM Brand. STMicroelectronics

GaN motor drive

As modern mechatronic systems demand smaller, more efficient motors, this GaN-based motor drive board offers a straightforward way to evaluate high-performance



motor control in a compact form factor. Integrated power components reduce losses, improve switching speed, and deliver smoother, more responsive motion compared with traditional designs.

The board supports real motor loads with both continuous and peak current capability and operates across a wide input voltage range. It supports both brushless and synchronous motors, offering flexibility across applications. Control, sensing, and protection features are fully integrated, including interfaces for programming, real-time debugging, and feedback for both open-loop and closed-loop control.

Applications. It is useful for robotics, drones, electric mobility,

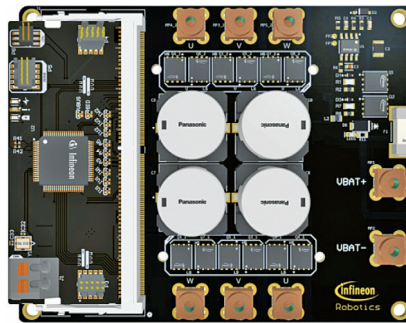
and other motion systems where space and efficiency matter.

Link. <https://www.electronicsforu.com/electronics-projects/compact-gan-based-inverter-reference-design>

OEM Brand. Efficient Power Conversion Corporation (EPC)

Dual-motor drive

When developing autonomous mobile robots, controlling two high-power BLDC wheel motors with limited processing resources can



be challenging, particularly when safety, efficiency, and ease of integration are critical.

This reference design addresses that challenge by demonstrating how a single controller can run field-oriented control for two motors while simultaneously managing sensing, communication, and protection. Safety is integrated through power management components that protect against high voltage, high current, and fault conditions, along with a high-side shutdown path for safe power cutoff when required.

System current is measured using a magnetic sensor that rejects stray magnetic fields and provides a stable analogue output to the controller. The design offers a practical example of reliable motor control, preparing robots for real-world conditions, such as slopes, uneven terrain, and rapid load changes.

Applications. This design is suitable for robotic applications

Link. <https://www.electronicsforu.com/electronics-projects/>

reference-design-dual-motor-drive-for-robotics

OEM Brand. Infineon

Autonomous robotics

This reference design provides a ready-made starting point for engineers developing autonomous mobile robots. It includes a complete hardware and software stack featuring a high-performance processor, an AI-capable GPU,



multiple cameras, ultrasonic sensors, and connectivity options, such as Wi-Fi, 4G, and 5G.

The platform enables you to manage sensing, perception, navigation, and communication while focusing on application development, robot mechanics, or specialised algorithms. It supports ROS 2 and Linux, operates reliably in demanding industrial environments, and accelerates the transition from concept to prototype while reducing integration risk.

Applications. This design is suitable for industrial and service robotics innovations.

Link. <https://www.electronicsforu.com/electronics-projects/reference-design-for-autonomous-robotics>

OEM Brand. Qualcomm

Together, these reference designs provide practical, ready-to-use starting points for building robots without reinventing the wheel. They accelerate prototyping and support faster, more confident development of next-generation autonomous robotic systems. **EFY**

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